

**THE SIXTEENTH WILLIAM LOWELL PUTNAM
MATHEMATICAL COMPETITION**

March 3, 1956

Morning Session

1. Evaluate

$$\lim_{x \rightarrow \infty} \left[\frac{1}{x} \frac{a^x - 1}{a - 1} \right]^{1/x}$$

where $a > 0, a \neq 1$. (page 418)

2. Prove that every positive integer has a multiple whose decimal representation involves all ten digits. (page 419)

3. A particle falls in a vertical plane from rest under the influence of gravity and a force perpendicular to and proportional to its velocity. Obtain the equations of the trajectory and identify the curve. (page 420)

4. Suppose the n times differentiable real function $f(x)$ has at least $n + 1$ distinct zeros in the closed interval $[a, b]$ and that the polynomial $P(z) \equiv z^n + C_{n-1}z^{n-1} + \cdots + C_0$ has only real zeros. Show that $(D^n + C_{n-1}D^{n-1} + \cdots + C_0)f(x)$ has at least one zero in the interval $[a, b]$ where D^n denotes, as usual, d^n/dx^n . (page 421)

5. Given n objects arranged in a row. A subset of these objects is called unfriendly if no two of its elements are consecutive. Show that the number of unfriendly subsets each having k elements is

$$\binom{n - k + 1}{k}. \quad \text{(page 422)}$$

6. (i) A transformation of the plane into itself preserves all rational distances. Prove that it preserves all distances.

(ii) Show that the corresponding theorem for the line is false. (page 423)

7. Prove that the number of odd binomial coefficients in any finite binomial expansion is a power of 2. (page 424)

Afternoon Session

1. Show that if the differential equation

$$M(x, y) dx + N(x, y) dy = 0$$

is both homogeneous and exact then the solution $y = f(x)$ satisfies $xM + yN = C$ (constant). (page 425)

2. Suppose that each set X of points in the plane has an associated set $\overline{X}$ of points called its cover. Suppose further that

(1) $\overline{X \cup Y} \supset \overline{X} \cup \overline{Y} \cup Y$, where $\cup$ designates point set sum (or union) and $\supset$ denotes set inclusion.

Prove: (i) $\overline{X} \supset X$, (ii) $\overline{\overline{X}} = \overline{X}$, (iii) $X \supset Y$ implies $\overline{X} \supset \overline{Y}$.

Prove conversely that (i), (ii) and (iii) imply (1). (page 426)

3. A sphere is inscribed in a tetrahedron and each point of contact of the sphere with the four faces is joined to the vertices of the face containing the point. Show that the four sets of three angles so formed are identical. (page 427)

4. Prove that if A , B , and C are angles of a triangle measured in radians then $A \cos B + \sin A \cos C > 0$. (page 428)

5. Consider a set of $2n$ points in space, $n > 1$. Suppose they are joined by at least $n^2 + 1$ segments. Show that at least one triangle is formed. Show that for each n it is possible to have $2n$ points joined by n^2 segments without any triangles being formed. (page 429)

6. Given $T_1 = 2$, $T_{n+1} = T_n^2 - T_n + 1$, $n > 0$, Prove:

(i) If $m \neq n$, T_m and T_n have no common factor greater than 1.

(ii) $\sum_{i=1}^{\infty} \frac{1}{T_i} = 1$. (page 430)

7. The polynomials $P(z)$ and $Q(z)$ with complex coefficients have the same set of numbers for their zeros but possibly different multiplicities. The same is true of the polynomials

$$P(z) + 1 \quad \text{and} \quad Q(z) + 1.$$

Prove that $P(z) \equiv Q(z)$. (page 431)